Received: 20 April 2023 | Accepted: 5 June 2023

DOI: 10.1002/ndr2.12187

New Disease Reports Open Access BSPP WILEY

NEW DISEASE REPORT

First report of target spot on tomato caused by *Corynespora cassiicola* in Nepal

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Funding information
Government of Nepal -Nepal Agricultural Research Council, Grant/Award Number: 194

KEYWORDS
Emerging disease, Nepal, Vegetable disease

Tomato (*Solanum lycopersicum*) is one of the most important vegetable crops in Nepal and is grown in high tunnels and open fields (Rawal et al., 2017). In June 2022, tomato plants having blighted lesions with concentric circles, yellow halos, and severely defoliated leaflets, with an incidence of c. 80%, were obtained from growers in the Panauti area of Kabrepalanchok district in the central hills of Nepal, an area of c. 3 ha. Samples were examined under a microscope with tape mounting and slightly curved to cylindrical, multiple pseudoseptate conidia resembling *Corynespora cassiicola* (Qi et al., 2011) were observed. Small pieces of infected leaves were surface disinfected with 1% sodium hypochlorite, washed twice with sterile water, and air-dried before being placed on acidified potato dextrose agar (PDA) plates for pathogen isolation. After five days incubation at 25°C with 12 hr light / 12 hr dark, grey to brown mycelia (Figure 1) with slightly curved cylindrical conidia were detected. Conidia (n = 30) were 105.8 μm (71.0- 142 μm) in length and 9.4 μm (7.2 -10.5 μm) in width, on average. The conidia were curved to straight or cylindrical in shape and formed in chains or singly (Figure 2). The specimen resembled *C. cassiicola* morphologically (Qi et al., 2011). Subsequently, pure cultures were obtained by culturing a single germinating spore on water agar plates before transferring to PDA plates.

DNA was extracted from week-old mycelia of pure cultures using AMPure XP beads (Beckman Coulter, UK) (Kumaishi et al., 2022). The internal transcribed spacer region of rDNA was amplified with the ITS4/ITS5 primers and sequenced. PCR amplifications were performed using the ITS4/ITS5 primer pair (White et al. 1990). PCR products were purified using AMPure XP beads before sequencing (Beijing Genomics Institute, Hongkong). The nucleotide sequences of two isolates RBK1 (GenBank Accession No. OQ504320) and RBK2 (OQ504321) were compared with sequences in GenBank using a BLASTN search, which resulted in 99.47% identity with *C. cassiicola* (FJ852574.1) isolated from tomato plants in the USA (Dixon et al., 2009; Sumabat et al., 2018).



FIGURE 1 Fungal colony incubated on potato dextrose agar after incubation at 25°C for seven days: top view (left), bottom view (right).

The pathogenicity of the fungal isolates was confirmed by inoculation of a spore suspension (1×10^4 conidia ml^{-1}) on ten 35-day old healthy tomato plants cv. Pusa Ruby in a greenhouse. At the same

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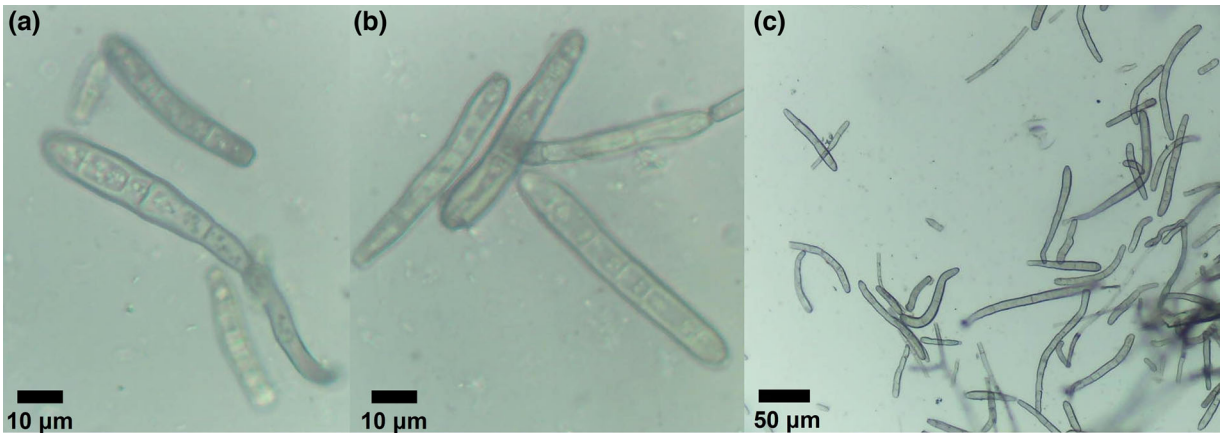


FIGURE 2 Conidia of *Corynespora cassiicola*.



FIGURE 3 Pathogenicity trial: a) tomato plants inoculated with *Corynespora cassiicola* showing symptoms three days post inoculation and b) non-inoculated plants.

time, ten healthy tomato plants were sprayed with sterile water to serve as negative control. The plants were maintained in greenhouse conditions, covered with a transparent plastic sheet, and misted frequently with sterile water to maintain humidity above 80% until 48 hours post-inoculation. Typical necrotic spots with characteristic target spot symptoms appeared in leaves and stems three days after inoculation and plants collapsed within two-three days after symptom development, but no such symptoms developed in non-inoculated plants during the 14 days the plants were monitored post-inoculation (Figure 3). The symptomatic leaves from inoculated plants and healthy leaves from non-inoculated plants were used for pathogen isolation as previously described. Isolates with same morphological characteristics as described above were recovered only from the leaves of the inoculated plants.

Based on the results of morphological, pathogenicity and sequence analyses, the fungal isolates were confirmed to be *C. cassiicola*. To the best of our knowledge, this is the first report of *C. cassiicola* in Nepal. Target spot is a major challenge for tomato production globally in tropical and subtropical regions (Schlub et al. 2009), indicating the pressing need for management strategies to be developed and increased awareness among farmers and extension workers to protect tomato plants from this emerging threat in Nepal.

ACKNOWLEDGEMENTS

The authors would like to thank Diane Saunders at John Innes Centre for providing AMPure XP beads. Authors also thank Bibek Dabargainya, Sachida Pokhrel, Shristi Upadhya, Arpan Pokhrel, Bibechana Poudel and Puspa Shrestha for support in the laboratory. The study was supported by the Government of Nepal through Nepal Agricultural Research Council.

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How to cite this article: Khadka, R.B., Bhandari, R., Rimal, A., Gaire, A., Adhikari, S., Bhandari, S. et al. (2023) First report of target spot on tomato caused by *Corynespora cassiicola* in Nepal. *New Disease Reports*, 48, e12187. https://doi.org/10.1002/ndr2.12187